NAME: ______________________________ **PERIOD:** ________ **DATE:** ______________

Which Tail, and Probability of What?

AP Statistics · Unit 3 · Topic 3.6 · B: 2026-12-04 · E: 2027-01-11

[STAR] TODAY'S PROBLEM

Suppose a city's earlier planning model says 55% of households support expanded evening bus service. A researcher suspects support has *fallen*. From the city's 20,000 households, a simple random sample of 200 finds 98 that support expansion, so $\hat{p} = 0.49$. Conditions for a one-sample z-test are met.

Priya: "For $H_0 : p = 0.55$ versus $H_a : p < 0.55$, use the left-tail p-value, about 0.044. If support is still 55%, about 4.4% of random samples of 200 would produce a sample proportion of 0.49 or lower."

Noah: "Any difference from 0.55 counts, so double the tail to about 0.088. That means there is an 8.8% chance that the actual support proportion differs from 0.55."

Sample counts

Group	Support	Other	Total
Households	98	102	200

FIRST TAKE — before the video (5 min)

Does fallen mean only a drop in support, or a change in either direction? Give one reason from the study.

[BOOK] A p-VALUE CONDITIONS ON THE NULL (keep on the page)

For a one-sample proportion test, $z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$. The p-value is the probability of a sample statistic **as extreme or more extreme** than the observed one, in the direction of H_a, assuming H_0 and the probability model are true. Use the left tail for $H_a : p < p_0$, the right tail for $H_a : p > p_0$, and both tails for $H_a : p \neq p_0$. A p-value is a probability about possible sample results under H_0; it is not the probability that a hypothesis is true.

Finish after the video:

DOK 1 **(a)** State the null and alternative hypotheses and identify the tail direction required by the research question.

DOK 2 **(b)** Compute the one-sample proportion z-statistic and its one-sided p-value. Also compute the two-sided p-value Noah would obtain.

DOK 3 **(c)** Choose the warranted p-value and interpretation. Use H_a and $\hat{p} = 0.49$ to define the tail event and null assumption, then explain what Noah's doubled area answers and what his statement gets wrong.

Frame: Because $H_a : p$______0.55, the p-value is the __________ tail beyond $\hat{p} =$ ________.

Frame: Assuming ____________________, a p-value of ________ is the probability that ____________________________________, not the probability that ________________________.

EXIT — turn this in

One thing the video changed about my first take: ____________________________________